

Communication Schemes for Centralized and Decentralized Event-Triggered Control Systems

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Abstract—Energy constraint long-range wireless sensor/actuator-based solutions are theoretically the perfect choice to support the next generation of city-scale cyber-physical systems. Traditional systems adopt periodic control which increases network congestion and actuations while burdens the energy consumption. Recent control theory studies overcome these problems by introducing aperiodic strategies, such as event-triggered control (ETC). In spite of the potential savings, these strategies assume actuator continuous listening, while ignoring the sensing energy costs. In this paper, we fill this gap, by enabling sensing and actuator listening duty cycling and proposing two innovative medium access control protocols for three decentralized ETC approaches. A laboratory experimental test bed, which emulates a smart water network, was modeled and extended to evaluate the impact of system parameters and the performance of each approach. Experimental results reveal the predominance of the decentralized ETC against the classic periodic control either in terms of communication or actuation by promising significant system lifetime extension.

Index Terms—Communication protocols, cyber-physical systems (CPSs), event-triggered control (ETC), networked control systems, wireless sensor/actuator networks.

I. INTRODUCTION

OVER the last decade, there has been a growing trend in industry to transform large-scale manual control and monitoring systems, such as electrical grids and water networks, into fully automatic cyber-physical systems (CPSs). The aim of this transformation is the improvement of quality of service (QoS) and the reduction of maintenance cost. In order to achieve these goals, plants and physical environments have

been augmented with sensor and actuator nodes which enable monitoring and control by communicating wirelessly and periodically to data centers or local base stations. However, these periodic dynamic control implementations introduce communication and energy consumption overheads.

In large scale CPS, sensor and actuator nodes are usually energy constraint and installed in harsh environments. For example, in smart water network more than 97% of sensing and actuation assets are located underground and powered by batteries [1]. To transmit the required information through long-range (several kilometres) wireless communications, high transmission power is required that leads to fast battery depletion. In addition, the periodic sensing, transmission, and actuation, regardless the state of the plant, decreases network bandwidth and increases actuations and consequently the energy consumption. Recent control theory studies propose to solve these problems by introducing aperiodic strategies, such as event-triggered control (ETC) strategies [2]–[7] in which the sensors and actuators communicate only when necessary.

In spite of the potential of significant savings, ETC techniques have only been partially examined and implemented on real systems, i.e., [8]–[19]. Eekelen *et al.* [10] propose a system based on the Diddyborg robot and examine the strategy presented in [7]. However, this system is first-order and, therefore, unable to test complex event-triggered strategies. In [19], an experimental evaluation was made for time-triggered control (TTC) and ETC from [2]. However, this paper requires state monitoring continuously to check event conditions. Additionally, the results in [2] can only be used for system with collocated sensors. To the best of our knowledge, there is no experiment that validates and compares different decentralized event-triggered mechanisms under the same conditions.

ETC systems have been studied extensively in order to guarantee convergence of plants under reduced communication schemes. However, the design and implementation of a communication protocol, which fully exploits the ETC behavior and ensures optimal communication, is still missing [20]. The state-of-the-art ETC approaches that are focused on communication, i.e., [21]–[29], have been limited to simulate or analyze theoretically the impact of network states on system performance. Carrier-Sense Multiple Access (CSMA) [30], Time-Division Multiple Access (TDMA) [31], and ALOHA [32] were the three communication protocols which have been used in the

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above-mentioned approaches. Specifically, [21], [22] provide useful insights and comparison of all the above-mentioned communication protocols. Ramesh *et al.* [23], Demirel *et al.* [24], and Ramesh *et al.* [25] present a Markov model that captures the joint interactions of the event-triggering policy and a contention resolution mechanism over CSMA communication. In [26] and [27], the ALOHA protocol, which has been applied in long term evolution random access procedure, was combined with ETC, with [27] to introduce the impact of collisions into the system performance. TDMA-based communication protocols, i.e., time-triggered network-on-a-chip and time-triggered controller area network, were analyzed in [28], which discusses their application to ETC systems. The earliest practical work is [19], continued in [29], by proposing the extension of the TDMA-based IEEE 802.15.4 Medium Access Control (MAC) layer [33] which has been used in communication protocols for network control via wireless, i.e., WirelessHART [34]. However, the main drawbacks of this approach are the assumption that the actuator nodes listen continuously to network messages. Furthermore, none of the prior work has considered the cost of sensing. For example, in our evaluation platform [35], the sensing costs almost the half the energy consumption of communication. In industrial systems with more energy hungry sensors, e.g., laser-based turbidity sensors for water quality, the energy cost may surpass communications.

Note that ETC is different than similar proposals on the communications domain [36]. Such event-based communications are data-driven techniques aimed at increasing also network energy efficiency. However, such techniques are oblivious of any requirements to guarantee stability in a real-time control application. In contrast, ETC approaches provide pre-designed stability and performance guarantees, while reducing resources' consumption.

Uniquely, in this paper, the proposed system avoids continuous sensing and state transmission from the sensor nodes, as well as radio listening for new control signals from the actuator nodes. These goals are achieved by applying periodic decentralized ETC strategies combined with innovative TDMA-based communication schemes.

Specifically, the contributions of this paper are listed as follows:

- 1) practical combination of sensor duty cycling with three decentralized aperiodic control approaches;
- 2) a new MAC protocol that facilitates decentralized synchronous ETC without the requirement of continuous actuator communication;
- 3) a novel flexible MAC protocol that can also accommodate two decentralized and asynchronous ETC approaches, communicating, first, absolute states and alternatively relative states only.

By using an extended version of the WaterBox test bed environment [35], we provide experimental results from TTC and four different ETC techniques: periodic centralized ETC (PETC) [4], periodic synchronous decentralized ETC (PSDETC) [5], and periodic asynchronous decentralized ETC (PADETC) by transmitting absolute or relative state [37]. To the best of our knowledge, this is the first real deployment

of most of the implemented ETC techniques to a real plant.

II. EVENT-TRIGGER CONTROL TECHNIQUES

We denote the positive real numbers by $\mathbb{R}^+$, the natural numbers including zero by $\mathbb{N}$. $|\cdot|$ denotes the Euclidean norm in the appropriate vector space, when applied to a matrix $|\cdot|$ denotes the l_2 induced matrix norm. A matrix $P \in \mathbb{R}^{n \times n}$ is said to be positive definite, denoted by $P \succ 0$, whenever $x^T P x > 0$ for all $x \neq 0$, $x \in \mathbb{R}^n$. For the sake of brevity, we write symmetric matrices of the form $\begin{bmatrix} A & B \\ B^T & C \end{bmatrix}$ as $\begin{bmatrix} A & B \\ \star & C \end{bmatrix}$.

A. Periodic Control

Consider a linear time-invariant plant [38] and controller

$$\dot{\zeta}(t) = A\zeta(t) + Bv(t) \quad (1)$$

$$v(t) = K\zeta(t) \quad (2)$$

where $\zeta(t) \in \mathbb{R}^n$ is the state vector and $v(t) \in \mathbb{R}^m$ is the input vector at time t . Assume $A + BK$ is Hurwitz, the system is completely observable, and each sensor can access only one of the system states.

A sample-and-hold mechanism is implemented for the controller (2)

$$v(t) = K\hat{\zeta}(t) \quad (3)$$

where

$$\hat{\zeta}(t) := \zeta(t_b), t \in [t_b, t_{b+1}[\quad (4)$$

and $\{t_b\}_{b \in \mathbb{N}}$ is the sequence of the update time of the state. Representing the sample-and-hold effect as a measurement error, we have

$$\varepsilon(t) := \hat{\zeta}(t) - \zeta(t). \quad (5)$$

Define T as the sample period. In a periodic TTC strategy, t_b is determined by

$$\{t_b | t_b = bT, b \in \mathbb{N}, T > 0\}. \quad (6)$$

B. Periodic Centralized Event-Triggered Control

In ETC strategies, the control input update time is determined by some pre-designed conditions. These conditions usually involve the system state and sample-and-hold error (5) [2]–[5]. Therefore, control executions happen only when necessary. However, the centralized event-triggered condition presented in [2] requires the continuous monitoring and transmission of the current state to check the event conditions. If the state cannot be measured continuously, we can either compute a stricter event condition considering measurement delays or apply the PETC strategy from [4], which combines periodic sampled-data control and ETC.

Consider system (1), (3), (5), and a sample sequence (6). At each sampling time t_b , the controller updates its state by

$$\hat{\zeta}(t_b) = \begin{cases} \zeta(t_b), & \text{when } \zeta_p^T(t_b) Q \zeta_p(t_b) > 0 \\ \hat{\zeta}(t_{b-1}), & \text{when } \zeta_p^T(t_b) Q \zeta_p(t_b) \leq 0 \end{cases} \quad (7)$$

where $\xi_p(t) = [\xi^T(t) \hat{\xi}^T(t)]^T$, Q satisfies $Q := \begin{bmatrix} (1-\sigma)I & -I \\ -I & I \end{bmatrix}$, and $\sigma > 0$.

For the system (1), (3), (5), and (6), if $\exists c > 0$ and $\rho > 0$ such that for any initial condition $\xi(0) \in \mathbb{R}^n$, $\forall t \in \mathbb{R}^+$, $|\xi(t)| \leq ce^{-\rho t}|\xi(0)|$ is satisfied, then the system is said to be globally exponential stable, we call ρ the decay rate [39].

According to [4, Corollary III.3], given a decay rate $\rho > 0$, if there exist a matrix $P > 0$ and scalars $\mu_i \geq 0$, $i \in \{1, 2\}$, such that

$$\begin{bmatrix} e^{-2\rho T} P + (-1)^i \mu_i Q & J_i^T e^{\bar{A}^T T} P \\ \star & P \end{bmatrix} > 0, \quad i \in \{1, 2\} \quad (8)$$

where

$$\bar{A} := \begin{bmatrix} A & BK \\ 0 & 0 \end{bmatrix}, \quad J_1 := \begin{bmatrix} I & 0 \\ I & 0 \end{bmatrix}, \quad J_2 := \begin{bmatrix} I & 0 \\ 0 & I \end{bmatrix}$$

then the system is globally exponential stable with a decay rate ρ .

C. Periodic Synchronous Decentralized Event-Triggered Control

The event-triggered strategies presented in (7) are centralized event-triggered strategies, since the event conditions require the whole vector of current state and current error. When the sensors are not colocated, decentralized event conditions are preferred. We introduce the PSDETC strategy based on [5].

For system (1), (3), and (5), a decentralized event-triggering condition with periodic sampling (6) is given by

$$\hat{\xi}(t_b) = \begin{cases} \xi(t_b), & \text{when } \exists i : \varepsilon_i^2(t_b) - \sigma \xi_i^2(t_b) > \theta_i \\ \hat{\xi}(t_{b-1}), & \text{when } \forall i : \varepsilon_i^2(t_b) - \sigma \xi_i^2(t_b) \leq \theta_i \end{cases} \quad (9)$$

where $\varepsilon_i(t)$ and $\xi_i(t)$ denote the i th coordinates of $\varepsilon(t)$ and $\xi(t)$, respectively, and $\{\theta_i\}_{i \leq n}$ is a set of parameters. Define $\{t_k\} := \{t_b | \exists i, \varepsilon_i^2(t_b) - \sigma \xi_i^2(t_b) > \theta_i\}$ the sequence of event times. The sequence $\{\theta_i\}_{i \leq n}$ is obtained solving at each event time t_k

$$\begin{cases} \hat{G}_i(t_k + t_e) = \hat{\varepsilon}_i^2(t_k + t_e) - \sigma \hat{\xi}_i^2(t_k + t_e) - \theta_i(k) \\ \hat{G}_i(t_k + t_e) = \hat{G}_j(t_k + t_e), \quad \forall i, j \in \{1, 2, \dots, n\} \\ \sum_{i=1}^n \theta_i(k) = 0 \end{cases} \quad (10)$$

where for $t \in [t_k, t_{k+1}]$

$$\begin{aligned} \hat{\xi}_i(t) &= \xi_i(t_k) + \dot{\xi}_i(t_k)(t - t_k) + \frac{1}{2}\ddot{\xi}_i(t_k)(t - t_k)^2 + \dots \\ &\quad + \frac{1}{q!}\xi_i^{(q)}(t_k)(t - t_k)^q \\ \hat{\varepsilon}_i(t) &= 0 - \dot{\xi}_i(t_k)(t - t_k) - \frac{1}{2}\ddot{\xi}_i(t_k)(t - t_k)^2 - \dots \\ &\quad - \frac{1}{q!}\xi_i^{(q)}(t_k)(t - t_k)^q. \end{aligned}$$

The map $t_e : \mathbb{N} \rightarrow \mathbb{R}^+$ can be set to either $t_e(k) = T$ or $t_e(k) = t_k - t_{k-1}$. We apply [5, Algorithm 1] to determine $t_e(k)$ in the experiments.

Thus, with the current $\theta_i(k)$ being calculated and transmitted from the controller to each sensor node, the sensor node can locally determine the occurrence of local events. When there is an event, the corresponding sensor node notifies the controller, and then the controller requests fresh measurements from all sensors to compute and update the control input.

Proposition 1: For system (1), (3), (5), and (9), given a decay rate $\rho > 0$, if there exist a matrix $P > 0$ and scalars $\mu_1, \mu_2, \mu_3 \geq 0$, such that

$$\begin{cases} \begin{bmatrix} e^{-2\rho T} P - \mu_1 Q & J_1^T e^{\bar{A}^T T} P \\ \star & P \end{bmatrix} > 0 \\ \begin{bmatrix} e^{-2\rho T} P + \mu_2 Q & J_2^T e^{\bar{A}^T T} P \\ \star & P \end{bmatrix} > 0 \\ \begin{bmatrix} e^{-2\rho T} P + \mu_3 Q & J_1^T e^{\bar{A}^T T} P \\ \star & P \end{bmatrix} > 0 \end{cases} \quad (11)$$

hold, then the system is globally exponential stable with a decay rate ρ .

Proof: According to [5], $\forall i : \varepsilon_i^2(t) - \sigma \xi_i^2(t) \leq \theta_i$ implies $\varepsilon^T(t)\varepsilon(t) \leq \sigma \xi^T(t)\xi(t)$, which is equivalent to $\xi_p^T(t_b)Q\xi_p(t_b) \leq 0$. However, $\exists i : \varepsilon_i^2(t) - \sigma \xi_i^2(t) > \theta_i$ may indicate $\xi_p^T(t_b)Q\xi_p(t_b) > 0$ or $\xi_p^T(t_b)Q\xi_p(t_b) \leq 0$. From the proof of [4, Corollary III.3], if the hypothesis in Proposition 1 holds, by applying the S-procedure [40], one obtains

$$\begin{aligned} x^T Q x > 0 \text{ holds, then } W(J_1 x, 0) &\leq W(x, T) \\ x^T Q x \leq 0 \text{ holds, then } W(J_2 x, 0) &\leq W(x, T) \\ x^T Q x \leq 0 \text{ holds, then } W(J_1 x, 0) &\leq W(x, T) \end{aligned}$$

in which $W(x, \tau)$ is a Lyapunov function defined as [4, eq. (18)]. Therefore, W does not increase during samplings. Together with the results from [4, Th. III.2], such a P can guarantee $\frac{d}{dt}W \leq -2\rho W$ between samplings; the system is globally exponential stable with a decay rate ρ . ■

D. Periodic Asynchronous Decentralized Event-Triggered Control

A periodic asynchronous ETC strategy is presented in [37]. In this strategy, again the triggering condition is distributed to each sensor node. When there is an event, in contrast with PSDETC, only the corresponding sensor node data are used to update the controller. The updated control input is then calculated and transmitted to the actuators.

Remark 1: In [37], the periodic asynchronous event-triggered mechanism requires **only** the transmission of the relative value of the state (i.e., increment and sign). In this paper, we call this mechanism PADETCrel. Additionally, we introduce the PADETCabs mechanism in which the nodes transmit the state (or absolute value) of the plant.

Consider system (1), (3), and (5), the current sampled state in the controller is updated as

$$\hat{\xi}_i(t_b) = \begin{cases} q(\xi_i(t_b)), & \text{if } i \in \mathcal{J} \\ \hat{\xi}_i(t_{b-1}), & \text{if } i \in \mathcal{J}_c \end{cases} \quad (12)$$

where $q(s)$ denotes a quantized signal of s , $\mathcal{J}$ is an index set: $\mathcal{J} \subseteq \{1, \dots, n\}$ for $\xi(t)$, indicating the occurrence of events, $\mathcal{J}_c := \{1, \dots, n\} \setminus \mathcal{J}$.

Define $\Gamma_{\mathcal{J}} := \text{diag}(\gamma_{\mathcal{J}}^1, \dots, \gamma_{\mathcal{J}}^n)$. The element $\gamma_{\mathcal{J}}^l$, with $l \in \{1, \dots, n\}$ is equal to 1, if $l \in \mathcal{J}$, and 0 otherwise. Furthermore, we use the notation $\Gamma_j = \Gamma_{\{j\}}$. The local event-triggering condition is

$$i \in \mathcal{J} \text{ iff } \xi_p^T(t_b) Q_i \xi_p(t_b) \geq \eta_i(t_b) \quad (13)$$

where $Q_i := \begin{bmatrix} \Gamma_i & -\Gamma_i \\ -\Gamma_i & \Gamma_i \end{bmatrix}$, $\eta_i(t) := \omega_i^2 \eta^2(t)$ is a local threshold, ω is a predesigned distributed parameter satisfying $|\omega| = 1$, $\eta(t)$ is a global threshold, determined by

$$\eta(t_b^+) = \begin{cases} \mu \eta(t_b), & \text{if } |\hat{\xi}(t_b^+)| \leq \varrho \eta(t_b) \wedge \eta(t_b) > \mu^{-1} \eta_{\min} \\ \eta_{\min}, & \text{if } |\hat{\xi}(t_b^+)| \leq \varrho \eta(t_b) \wedge \eta(t_b) \leq \mu^{-1} \eta_{\min} \\ \mu^{-1} \eta(t_b), & \text{if } |\hat{\xi}(t_b^+)| \geq \mu^{-1} \varrho \eta(t_b) \\ \eta(t_b), & \text{otherwise} \end{cases} \quad (14)$$

where $\eta_{\min} > 0$ is a prespecified minimum threshold, $\mu \in]0, 1[$ is a predesigned parameter, and ϱ is a design parameter which can be determined by validating the feasibility of some bilinear matrix inequalities (17).

Consider the Hamiltonian matrix

$$H := \begin{bmatrix} H_{11} & H_{12} \\ H_{21} & H_{22} \end{bmatrix} \quad (15)$$

where $H_{11} := \bar{A} + \rho I$, $H_{12} := 0$, $H_{21} := -(\gamma^2 I - I)^{-1}$, $H_{22} := -(\bar{A} + \rho I)^T$, for some $\gamma > 1$, and some given $\rho > 0$. And introduce the matrix exponential

$$F(\tau) := e^{-H\tau} = \begin{bmatrix} F_{11}(\tau) & F_{12}(\tau) \\ F_{21}(\tau) & F_{22}(\tau) \end{bmatrix}. \quad (16)$$

Assumption 1: $F_{11}(\tau)$ is invertible $\forall \tau \in [0, T]$.

Base on [4], if Assumption 1 holds, then the matrix $-F_{11}^{-1}(T)F_{12}(T)$ can be guaranteed to be positive semidefinite. Define the matrix $\bar{S}$ satisfying $\bar{S}\bar{S}^T := -F_{11}^{-1}(T)F_{12}(T)$.

According to [37, Th. IV.4], consider the system (1), (3), (5), (12), and (14), given the Assumption 1 holds and a decay rate $\rho > 0$. If there exist matrix $P > 0$, scalars $\varrho > 0$, $\beta_1 > 0$, $\beta_2 > 0$, and $\epsilon_{\mathcal{J}_i} > 0$, $\mathcal{J} \subseteq \{1, \dots, n\}$, $i \in \{1, \dots, n\}$ such that the bilinear matrix inequality

$$\begin{bmatrix} \beta_2 I & F_{11}^{-T}(T)P\bar{S} & \tilde{F} & -\beta_2 J_{\mathcal{J}} & 0 \\ * & I - \bar{S}^T P \bar{S} & 0 & 0 & 0 \\ * & * & \tilde{F} & 0 & 0 \\ * & * & * & P + \tilde{H}_1 & 0 \\ * & * & * & * & \tilde{H}_2 \end{bmatrix} \succ 0 \quad (17)$$

holds, where

$$\begin{aligned} \tilde{F} &:= F_{11}^{-T}(T)P F_{11}^{-1}(T) + F_{21}(T)F_{11}^{-1}(T) \\ \tilde{H}_1 &:= -\beta_1 I + \beta_2 J_{\mathcal{J}}^T J_{\mathcal{J}} - \sum_{i \in \mathcal{J}} \epsilon_{\mathcal{J}_i} Q_i + \sum_{i \in \mathcal{J}_c} \epsilon_{\mathcal{J}_i} Q_i \\ \tilde{H}_2 &:= \beta_1 \varrho^2 I - \beta_2 \bar{\Delta}_{\mathcal{J}}^T \bar{\Delta}_{\mathcal{J}} + \sum_{i \in \mathcal{J}} \epsilon_{\mathcal{J}_i} \Theta^T \Gamma_i \Theta \\ &\quad - \sum_{i \in \mathcal{J}_c} \epsilon_{\mathcal{J}_i} \Theta^T \Gamma_i \Theta \\ \bar{\Delta}_{\mathcal{J}} &:= \begin{bmatrix} 0 \\ \Gamma_{\mathcal{J}} \Theta \end{bmatrix}, \quad J_{\mathcal{J}} = \begin{bmatrix} I & 0 \\ \Gamma_{\mathcal{J}} & I - \Gamma_{\mathcal{J}} \end{bmatrix} \\ \Theta &= [\omega_1 \cdots \omega_n]^T \end{aligned}$$

then $\mathcal{A} := \{x | |x| \leq \bar{\varrho} \eta_{\min}\}$ is a globally preasymptotically stable set for the system [41], where $\bar{\varrho} := \max\{|J_{\mathcal{J}}| \varrho + |\bar{\Delta}_{\mathcal{J}}|, \forall \mathcal{J}\}$. Intuitively, the globally preasymptotically stability indicates that the hybrid system state converges to a set along the hybrid timeline. This set is the globally preasymptotically stable set.

In [37], the update of the signals $\hat{\xi}_i(t_b)$ is given by

$$\hat{\xi}_i(t_b) = \hat{\xi}_i(t_{b-1}) + \text{sign}(\hat{\xi}_i(t_{b-1}) - \xi_i(t_b)) m_i(t_b) \sqrt{\eta_i(t_b)} \quad (18)$$

in which $m_i(t_b) := \lfloor (|\hat{\xi}_i(t_{b-1}) - \xi_i(t_b)|) / (\sqrt{\eta_i(t_b)}) \rfloor$, $i \in \{1, \dots, n\}$. Thus, in practice, the $\text{sign}(\hat{\xi}_i(t_{b-1}) - \xi_i(t_b))$ and $m_i(t_b)$ has to be transmitted from the sensor to the controller. The increment and sign represent the relative value in PADETCrel. In PADETCabs, the signals $\hat{\xi}_i(t_b)$ is instead updated by $\hat{\xi}_i(t_b) = \xi_i(t_b)$. Note that, all the ETC mechanisms have a fine static quantizer due to the analog to digital conversion. In PADETCrel, the sensor nodes transmit only the relative state. Therefore, an additional coarse dynamic quantizer can be applied with maximum quantization error $\sqrt{\eta_i(t_b)}$ for each sensor, i.e., let $q(\xi_i(t))$ in (12) be of the form shown in (18). However, since the fine static quantizers we install in our experimental setting have quantization error negligible compared with the noise, only PADETCrel considers the output quantization error.

III. INCORPORATING ETC WITH THE MAC LAYER

In this section, we present the design and implementation of three innovative TDMA-based MAC protocols which enable the deployment of TTC, PETC, PSDETC, and PADETC approaches accordingly: control-TDMA (C-TDMA), synchronous decentralized C-TDMA (SDC-TDMA), and asynchronous decentralized C-TDMA (ADC-TDMA). The main benefits of these ETC-specific MAC protocols are: the optimization of communications by fully exploiting the behavior and needs of ETC; the minimization of actuator node listening through duty cycling; and the off-load of the local controller node by allowing **only** one node to communicate with a base station per time.

For the proposed TDMA-based communication schemes, we assume a city-scale CPS network infrastructure, such as in Fig. 1, which represents a smart water network. In this architecture, the sensors and actuators are divided into clusters.

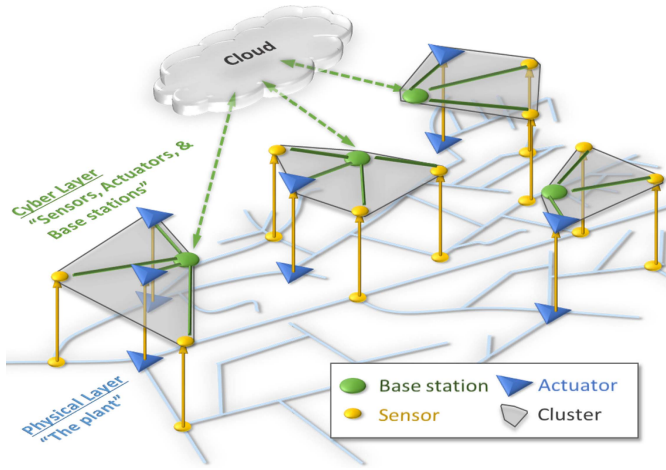


Fig. 1. Network architecture.

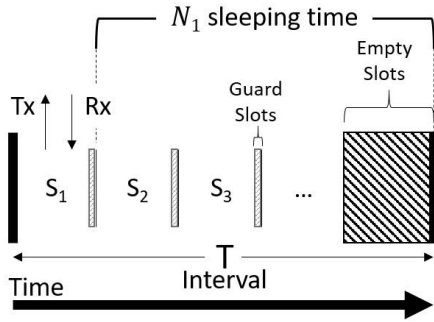


Fig. 2. Simplistic TDMA MAC protocol.

Each clusters consists of only the sensors, actuators, and a base station which are involved in the control loop of an application for a specific area. The sensor/actuator nodes communicate in single-hop¹ with a base station and retrieve acknowledgment messages per transmission. Based on this architecture, we assume a star topology in which all the nodes of the same cluster can communicate with the base station (in which the controller is running) of the cluster. Contemporary long-range communication technologies, such as low-power wide-area [42], are representative examples of embodiments this architecture. Furthermore, the controller, which computes the control signals, is executed in the base station. In this paper, the terms controller and base station are used interchangeably. Our communication schemes are applied within each cluster while the information exchange among different clusters is out of the scope of this paper. Note that, despite the single-hop and centralized communication, the triggering of state transmission is distributed for decentralized ETC mechanisms.

A. Simplistic TDMA Protocol

TDMA is a channel access method for shared medium networks, which allows several users to share the same frequency channel. Specifically, time is divided into intervals T_i with length T , so-called superframes. Each interval is split into smaller time slots S_j , with $\sum_{j=1}^N S_j \leq T$, where N is

¹We selected a single-hop communication, because a dynamic multihop network infrastructure cannot provide guarantees for time delays; a critical factor for control systems.

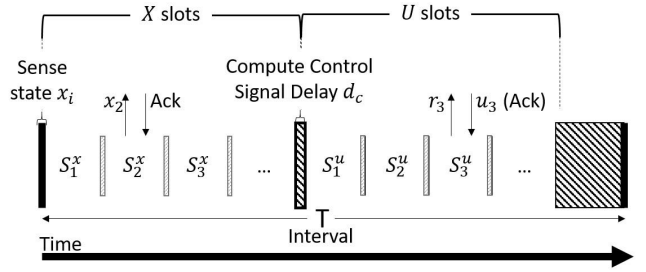


Fig. 3. CTDMA protocol.

the number of sensor/actuator nodes which share the same channel.² In each time slot, only one predefined sensor/actuator node N_j can transmit (T_x) or receive (R_x) a burst of messages to and from a base station. Outside the timeslot S_j , N_j sleeps or executes other tasks depending on the hardware infrastructure and the provided ability to duty cycle. To avoid time violations of time slot bounds due to N_j possible clock drift, a guard slot forces the termination of communication before the end of each S_j . Fig. 2 shows the communication scheme of a simplistic TDMA protocol.

Due to synchronous operation, a TDMA-based protocol [34] can guarantee tight bounds on delays which are critical for network control systems. On the other hand, synchronizing sensor/actuator nodes is considered as the main drawback of TDMA-based systems. However, the state-of-the-art solutions, i.e., GPS clock synchronization technologies [43], ensure typical accuracy better than $1 \mu s$ by consuming ultralow power and without introducing communication overheads. This time synchronization technology has been tested widely, in term of robustness and performance, in real city-scale deployments, such as the smart water network in [44].

B. C-TDMA, and TTC and PETC

C-TDMA is designed to enable TTC and PETC approaches (see Fig. 3). In order to ensure the simultaneous state sampling, in the beginning of every interval T_i at time t_i , every node N_j has to retrieve a state measurement x_j from the available sensors. Then, the channel bandwidth is divided into two sets of time slots which are separated by a time delay.

1) *Measurement Slots S_j^x (X-Slots)*: Every sensor/actuator node N_j transmits x_j within the time slot S_j^x to the controller. Within each time slot only one successful message is required. Thus, the size selection of S_j^x is application specific and depends on the size of x_j (e.g., 2 Bytes per sensor) and the number of retransmissions to achieve high reliability based on the selected hardware.

2) *Delay d_c* : After receiving the complete sampled state by receiving $x_j, \forall j \in N$, a time delay is required to allow the computation of appropriate control signals u_j for every sensor/actuator node N_j . The length of this delay depends on the controller infrastructure and the complexity of the control model.

²A superframe can be divided into equal time slots that fully utilize the channel or to minimal slots which cover the application requirements and allow the communication to new nodes into the same channel.

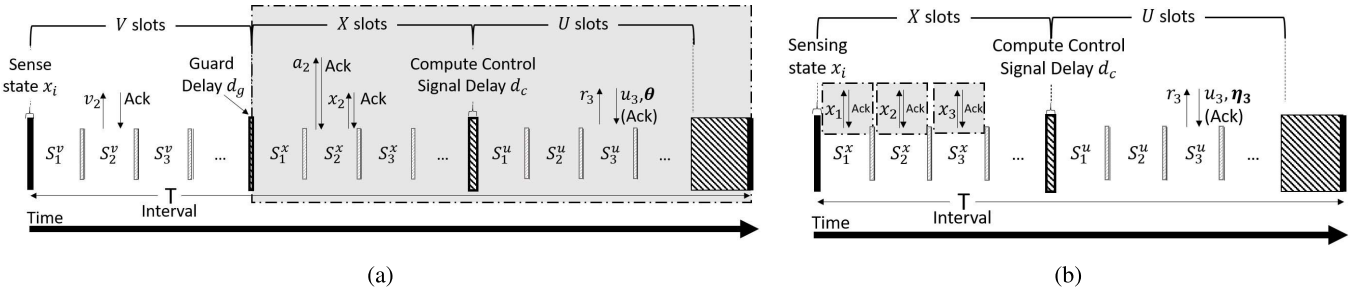


Fig. 4. Communication schemes for decentralized ETC techniques. (a) SDC-TDMA protocol for PSDETC approach. (b) ADC-TDMA protocol for PADETC approach.

3) *Actuation Slots S_j^u (U-Slots)*: The last set of time slots is related to the control message retrieval by the sensor/actuator nodes N_j . In each time slot S_j^u , node N_j transmits a request r_j for a control signal u_j to the controller. Then, u_j is piggybacked to the acknowledgment message. The benefit of r_j request is twofold: 1) off-loads the controller side and 2) reduces N_j listening time. Otherwise, the controller has to transmit or broadcast u_j continuously by blocking other tasks, while N_j has to be active in receive mode during the full length of the S_j^u slot until a successful control message retrieval. Furthermore, this approach causes more energy savings for nodes with communication modules that consume the same amount of energy for transmission and listening, i.e., [45]. The length of S_j^u depends on the size of u_j signals.

Based on the above, the minimum interval size T_{min} can be defined by the length of X-slots, U-slots, and delay d_c , and the number of the nodes. Furthermore, T_{min} can be considered as the maximum time delay of the system.

TTC and PETC are centralized approaches and are executed in the controller. In both cases, the system requires the transmission of the current state to the controller at every T_i during the X-slots. Then, in the TTC technique, the controller replies back in the U-slots of every interval T_i with a new u_j control signal. On the other hand, in PETC, the controller evaluates the event condition, as has been described in (7), and transmits the new u_j only if there is a violation. This behavior allows PETC to save energy due to actuation reduction. Note that the base station has only as extra overhead the evaluation of the event-triggered condition in (7). Therefore, the introduced computational complexity is minor compared with that of TTC systems.

C. SDC-TDMA and PSDETC

PSDETC is a distributed technique and each sensor node is responsible for the state transmission decision in every T_i . The computation of control signals u_j and θ_j parameters require the **complete** knowledge of the system's state for the interval T_i in which a threshold violation has happened. For example, consider a system with three nodes, $\{N_1, N_2, N_3\}$, in which only two of the nodes, i.e., $\{N_2, N_3\}$, observe threshold violations. The controller requires the state from all the three nodes to compute the control signals. Using the same example, in a TDMA-based communication scheme in which each node is assigned to a predefined time slot S_j , node N_1 is precluded from transmitting its

state by being unable to be informed about the threshold violations of N_2 and N_3 . To overcome these limitations, SDC-TDMA introduces a new set of time slots S_j^v , the V-slots [see Fig. 4(a)].

1) *Violations Slots S_j^v (V-Slots)*: In the beginning of every T_i , each node retrieves a measurement and evaluates (9). Then, the result of each threshold v_j is transmitted by the corresponding node N_j to the controller at time slot S_j^v .

2) *Measurement Slots S_j^x (X-Slots)*: In the beginning of every S_j^x in X-slots, each node N_j asks the controller, by sending an a_j request, whether a threshold violation was observed in the V-slots. If no threshold violation occurred, the sensor node sleeps immediately until the next interval T_{i+1} [gray box in Fig. 4(a)]. Otherwise, each node transmits each state x_j to controller, wait for the delay d_c and actuation slots, U-slots, follow.

3) *Delay d_c and Actuation Slots S_j^u (U-Slots)*: Similar to the C-TDMA approach, after a delay d_c , each node requests the new control signal u_j from the controller. u_j and the new threshold parameters θ_j , which is being calculated based on (10), are being piggybacked into the acknowledgment messages of U-slots.

Based on the above, SDC-TDMA sacrifices channel availability and increases the minimum interval length, T_{min} and consequently the system's maximum delay, by adapting V-slots into the TDMA scheme. However, in the case of threshold violation absence, the communication is being minimized significantly by avoiding the transmission of state x_j and the entire execution of U-slots.

D. ADC-TDMA and PADETC

Similar to PSDETC, the PADETC approach transfers the communication decision making from the controller down to the sensor/actuator nodes. Additionally, due to its asynchronous feature, this approach increases the communication savings by avoiding the state transmission x_j from every node N_j in every interval T_i . The only overhead in the communication is η_j update based on (14) and transmission to N_j . The value of η_j is being piggybacked with the control signal u_j in the acknowledgment message.

Specifically, the architecture of ADC-TDMA is similar to C-TDMA and consists of sensing task, X-slots, d_c delay, and U-slot [see Fig. 4(b)]. In a S_j^x slot, the node N_j evaluates the threshold of (13). In the case of no threshold violation, the node N_j skips the communication and returns to sleep

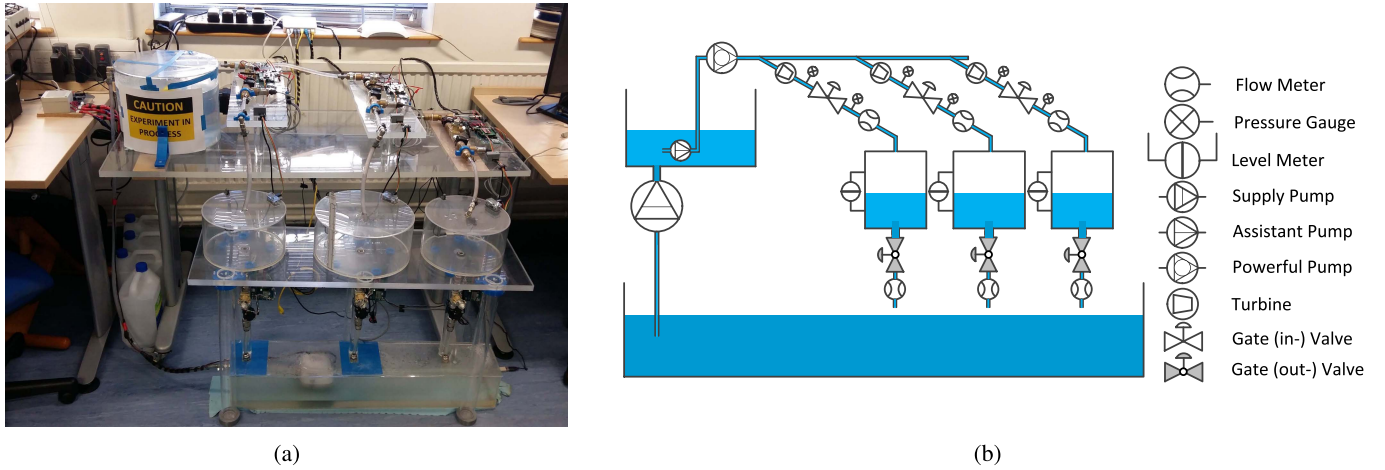


Fig. 5. WaterBox evaluation platform. (a) WaterBox test bed. (b) Schematic of the WaterBox.

mode until S_j^u . For example, the communication in gray boxes of Fig. 4(b) can be avoided completely or partially depending on the violations. Otherwise, N_j transmits to the controller: x_j in PADETCabs or the increment m_j in PADETCrel (see Section II-D). Then, the controller computes the appropriate control signals and updates the local and global η based on (14) by using **only** the available x_j states.

In the U-slots, every node has to send a r_j request message to the controller, in order to be notified about a possible threshold violation from another node. Therefore, the violation of at least one threshold causes the update of u_j and η_j to be sent to every actuator node. The values of u_j and η_j are piggybacked to the acknowledgment message.

IV. EVALUATION PLATFORM: WATERBOX

Smart water networks have been used as a proof of concept for our proposed framework. The WaterBox platform (see Fig. 5) is a scaled version of such a water network [35] and developed to demonstrate real time monitoring and control by adapting innovative communication theories and control methodologies.

In this paper, our aim is to demonstrate the event-triggered techniques on possible practical relative setups. Therefore, WaterBox was used as evaluation platform for our proposed ETC techniques and communication schemes. In the future, the same infrastructure will allow us to evaluate nonlinear event-triggered mechanisms.

A. WaterBox Infrastructure

A Water supply network structure consists of three individual layers: 1) storage and pumping; 2) water supply zones and district meter areas (DMAs); and 3) end users (water demand). While valves control flows and pressures at fixed points in the water network, pumps pressurise water to overcome gravity and frictional losses along supply zones, which are divided into smaller fixed network topologies (in average 1500 customer connections) with permanent boundaries, DMAs. The DMAs are continuously monitored with the aim to enable proactive leakage management, simplistic pressure management, and

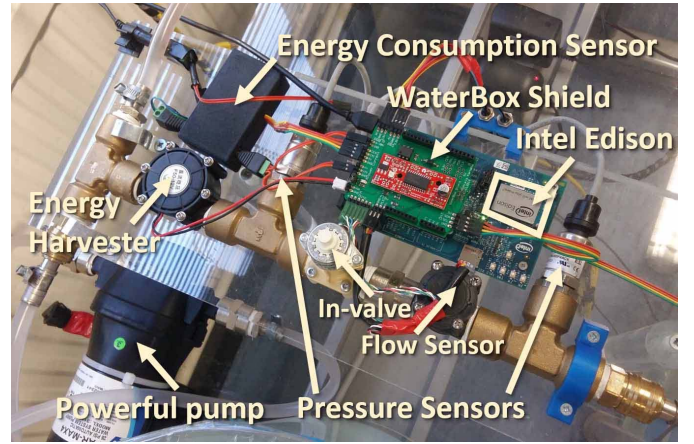


Fig. 6. WaterBox sensor/actuator node.

efficient network maintenance. WaterBox was designed to support this architecture as follows.

1) *Water Storage and Pumping*: The structure of the WaterBox is shown as Fig. 5(b). The WaterBox has a lower tank (i.e., ground and soil), an upper tank (i.e., reservoir and lake), and three small tanks (i.e., DMAs). The lower tank collects water from small tanks, and supplies water to the upper tank by an underwater bilge supply pump. This supply pump can supply enough water as the system requires. An assistant bilge pump and a new powerful pump are installed in series inside and after the upper tank, respectively, and supply water to the small tanks. When the small tanks require more water supply, the assistant pump and powerful pump work together. When the small tanks require less water supply, only the assistant weak pump works. This behavior emulates the day and night pumping operation of a water network in which the demand changes dramatically.

2) *Water Supply and Sensor/Actuator Node*: The water from the powerful pump flows into three small “DMA” tanks via a pipe network. For the inlet each tank, a sensor/actuator node (see Fig. 6), based on the Intel Edison development board [46], controls the water flow through a motorized gate valve, so-called in-valve, and monitors the water flow,

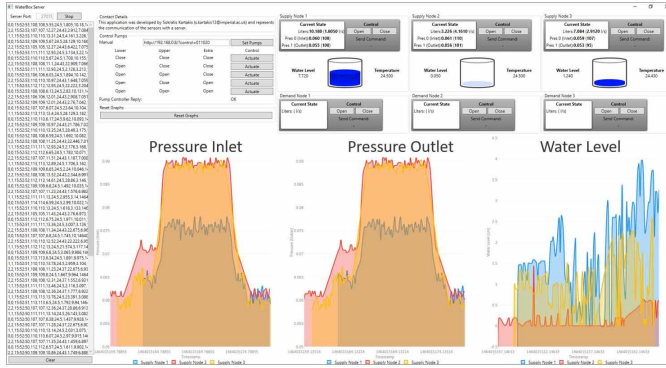


Fig. 7. Local controller visualization application.

pressure (before and after in-valve), and the in-tank water level. Furthermore, a turbine (flow-based energy harvester) is installed before each in-valve to harvest energy. To capture the total energy consumption of each sensor/actuator node, a custom made sensor module was created.

Remark 2: In the WaterBox infrastructure, the sensors and actuator of each inlet are connected to the same node. However, our proposed communication schemes can be applied to noncollocated infrastructures.

3) *Water Demand Emulation:* At the bottom of each small tank, there is an opening which enables the emulation of water consumption. A gate valve, so-called out-valve, is installed after each opening and can be controlled by a sensor/actuator node. The control of out-valves changes the outlet flow rate and facilitates the emulation of user's random water consumptions (i.e., external disturbance).

4) *Base Station (Controller):* Every sensor node is connected to a local isolated Wi-Fi network.³ A laptop is used as a controller or base station and hosts a visualization application (see Fig. 7) which presents at real time the current state of the system, allows the manual control of actuators, logs the retrieved messages, and enables our proposed communication schemes and ETC scenarios per experiment. Additionally, due to lack of an indoor GPS time synchronization, an Network Time Protocol (NTP) server is running in the local controller and ensures less than millisecond time synchronization accuracy among the nodes. To avoid the communication overhead of the NTP approach, each node executes the synchronization process only at the beginning of each experiment.

B. System Identification and Modeling

We apply gray-box identification [47] to generate the system model and to find the uncertain parameters. A first principles model is obtained following [48]. We identify independently models under both mode 1: only the assistant pump works and mode 2: both pumps work. Using the MATLAB cftool, we generate fitting curves for the gate valve coefficient of each in-valve, the turbine efficiency, and the pump efficiency. These

³The isolation was achieved by disabling service set identifier operation (broadcast of Wi-Fi availability to new users) from the router and selecting the low occupied communication channel for the nodes based on spectrometer experiments

curves are used to compute the first principles model. Since our aim is to stabilize the water level of each tank $j \in \{1, 2, 3\}$ at the desired height h'_j , the model is linearized around this height. In this process, in order to simplify the simulation of the user water consumption, we keep the out-valves open, thus, constant out flow rates can be assumed.

V. HYBRID CONTROLLER DESIGN

To evaluate the proposed ETC-based communication schemes, the following **control scenario** was used: “Control in-valves to stabilize the water level of the small ‘DMA’ tanks to a certain level ensuring pressure and flow bounds. Enable mode 2 (weak and powerful pump), only if the system is away from the target levels. Switch to mode 1 (weak pump), only when the system is close to the reference state to guarantee efficient low pressure and flow operation.”

In the design of the controller, the following limitations need to be considered.

- 1) *Saturation of the Actuators:* The maximum open level of one in-valve is 360° , while the minimum is 0° .
- 2) *Actuator Quantization:* Due to the limitation of the valve's control components, their open levels can only be changed in steps of 10° . Therefore, small disturbances may result in dramatic changes of actuators.
- 3) *Over-Pressure Protection:* Due to mechanical limitations of the pipe network, there is a maximum allowable pressure for the pipe network.

Since the height of the water levels have a direct effect on the QoS, the closed-loop system requires a fast response; however, since the size of the small tanks are limited, the overshoot should simultaneously be constrained. Experiments show that, in mode 2, the pipe network may be over pressured, when the open level of the in-valves, defined by α_j^{in} , cannot satisfy

$$\sum_{j=1}^3 \alpha_j^{in} \geq 180^\circ. \quad (19)$$

Overshoot and disturbances could make condition (19) be violated. While in mode 1, there is no such constrains, that is, even all three in-valves are totally closed, the pipe network will not be over pressured. Therefore, filling the small tanks in mode 2 and switching to mode 1 when (19) is violated is required. Also experimentally, we observe that, when the system is in mode 1, the pump may not provide enough water supply to the small tanks even at the maximum open level, i.e., $\alpha_j^{in} = 360^\circ$, $\forall j \in \{1, 2, 3\}$. Therefore, switching back to mode 2 when the water in the tanks reaches some predefined low levels is necessary. To support this mode switching, we define $\underline{h} := [\underline{h}_1 \ \underline{h}_2 \ \underline{h}_3]^T$, $\underline{h}_j < h'_j$, $j \in \{1, 2, 3\}$, as the lower water levels. If $\exists j \in \{1, 2, 3\}$ such that $h_j(t) \leq \underline{h}_j$, the system switches from mode 1 to 2. With carefully chosen $\underline{h}_j$ and properly designed controllers, this violation can only happen in mode 1. Further analysis shows that, (19) can only be violated when $h_j(t) > h'_j$, which together with the fact that $\underline{h}_j < h'_j$ precludes Zeno behavior.

Let $\vartheta \in \{1, 2\}$ represents the corresponding system mode. The linearized switched model and switched controller of

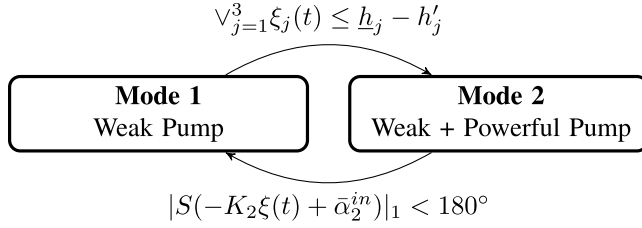


Fig. 8. Hybrid controller state automaton.

WaterBox are described by

$$\dot{\xi}(t) = A(\xi(t) + h') + B_\vartheta v(t), \quad \vartheta = 1, 2 \quad (20)$$

$$v(t) = S(-K_\vartheta \xi(t) + \bar{\alpha}_\vartheta^{in}), \quad \vartheta = 1, 2 \quad (21)$$

where $\xi(t) = [\xi_1(t) \ \xi_2(t) \ \xi_3(t)]^T$, $\xi_j(t) := h_j(t) - h'_j$, $j = 1, 2, 3$ are the system states, $h_j(t) \in \mathbb{R}$ are the water levels, and $h'_j \in \mathbb{R}$ are the reference water levels with $h' = [h'_1 \ h'_2 \ h'_3]^T$; $v(t) = [v_1(t) \ v_2(t) \ v_3(t)]^T$, $v_j(t) := \alpha_j^{in}$, $j = 1, 2, 3$ are the system control inputs, and $\bar{\alpha}_\vartheta^{in}$ are the equilibrium open levels of the in-valves per operation mode ϑ ; S is a map $\mathbb{R}^m \rightarrow \mathbb{R}^m$ representing actuator saturation and quantization, that is $S(s_j) := \max\{\min\{10\lfloor 10^{-1}s_j \rfloor, 360^\circ\}, 0^\circ\}$.

Then, the WaterBox hybrid controller state automaton is shown in Fig. 8.

From Gray-Box identification procedure, the system parameters are defined as follows. Due to the physical structure of the tanks and the low sensitivity of the flow meters, the flow rates are identified as constants, that is

$$A(\xi(t) + h') = -10^{-4} \times [5.809 \quad 3.554 \quad 5.102]^T.$$

And

$$B_1 = 10^{-5} \times \begin{bmatrix} 0.1436 & -0.0170 & -0.0164 \\ -0.0098 & 0.1060 & -0.0100 \\ -0.0139 & -0.0139 & 0.1492 \end{bmatrix}$$

$$B_2 = 10^{-5} \times \begin{bmatrix} 0.7666 & -0.0493 & -0.0457 \\ -0.0274 & 0.5848 & -0.0279 \\ -0.0393 & -0.0432 & 0.7865 \end{bmatrix}.$$

The controllers designed are given by

$$K_1 = \begin{bmatrix} 99950 & 3029 & 872 \\ -3014 & 99940 & -1679 \\ -922 & 1652 & 99982 \end{bmatrix}$$

$$K_2 = \begin{bmatrix} 9998.5 & 167.1 & 41.0 \\ -166.6 & 9997.9 & -116.0 \\ -43.0 & 115.3 & 9999.2 \end{bmatrix}.$$

The designed controller is stable in both modes 1 and 2, because $-B_1 K_1$ and $-B_2 K_2$ are Hurwitz matrices. Furthermore, due to the long dwell time of the system, the closed loop retains stable. Given $h'_j = 0.06$ and $\underline{h}_j = 0.03$, $\forall j \in \{1, 2, 3\}$, $\bar{\alpha}_1^{in}$ and $\bar{\alpha}_2^{in}$ are computed by solving $A(\xi(t) + h') + B_\vartheta \bar{\alpha}_\vartheta^{in} = 0$

$$\bar{\alpha}_1^{in} = \begin{bmatrix} 503.5950 \\ 422.4378 \\ 428.5839 \end{bmatrix}, \quad \bar{\alpha}_2^{in} = \begin{bmatrix} 84.5099 \\ 68.2069 \\ 72.8442 \end{bmatrix}.$$

Since $A(\xi(t) + h')$ is compensated by $B_\vartheta \bar{\alpha}_\vartheta^{in}$, A in (1) can be treated as a zero 3×3 matrix. For the presented event-triggered

TABLE I
COMMUNICATION PARAMETER EVALUATION SETUP

	Parameter	Value	Description
Packet Size	x_j	36 Bytes	Node ID Timestamp (in msec) Inlet pressure Outlet pressure Flow rate Total water volume Distance from surface Energy consumption Energy harvesting
	$Ack - r_j, a_j$	1 Byte	0 or 1 — Node ID
	u_j, η_j, θ_j	2 Bytes	Control signal and DETC parameters
	m_j	4 Bytes	State delta
Time Duration	S_j^x	80 msec	X slot size
	S_j^u, S_j^v	50 msec	U and V slot size
	d_c	10 msec	Control decision delay
	d_g	5 msec	Threshold violation decision delay
	Guard slot	1 msec	Forced task termination time

mechanisms, once the inequalities shown in (8), (11), and (17) are feasible, the system is stable either if $A = 0$ or $A \neq 0$. However, the specific dynamics may not result in a large difference regarding the effects of ETC on communications. In our future work, further analysis and tests will be conducted with different system dynamics for more conclusive results. Note that, since the linearized model, switching condition, and input quantization are the same for all the event-triggered mechanisms, the comparison is fair.

VI. EVALUATION

This section summarizes the experimental results of more than 400 experiments conducted in WaterBox to evaluate our proposed communication schemes for the different ETC strategies. Each experiment executes the same control scenario (as described in Section V) and the total process lasts between 7 and 10 min, including the water state initialization, the execution of experiment, and data logging from sensor/actuator nodes and local controller.

A. Evaluation Setup

The proposed communication protocols of Section III were deployed to the WaterBox sensor/actuator nodes by wrapping the functionality of the Intel Edison Wi-Fi module. Table I presents the configuration of the communication parameters. Based on the predefined packet sizes of the specific hardware infrastructure, a set of experiments was conducted to determine reliable time slot and guard delay sizes.

Based on Table I timing parameters and Section III time slot analysis, the minimum interval size T_{min} for C-TDMA and ADC-TDMA has to be more than 406 ms while for SDC-TDMA more than 564 ms (because of the V-slots). Thus, we evaluate TTC, PETC, and PADETC (with absolute or reference value) with interval size $T = 0.5, 1, 2$ and $T = 1, 2$ s for PSDETC. The selected interval sizes and the rest parameters of the ETC strategies are listed in Table II. The σ and ρ

TABLE II
PARAMETERS OF TRIGGERING STRATEGIES

Method	T (sec)	Parameter	Value
TTC	0.5, 1, 2	-	-
PETC	0.5, 1, 2	σ	0.05, 0.1, 0.2
PSDETC	1, 2		
PADETC (abs & rel)	0.5, 1, 2	μ	0.75, 0.95
		ϱ	85 , 120

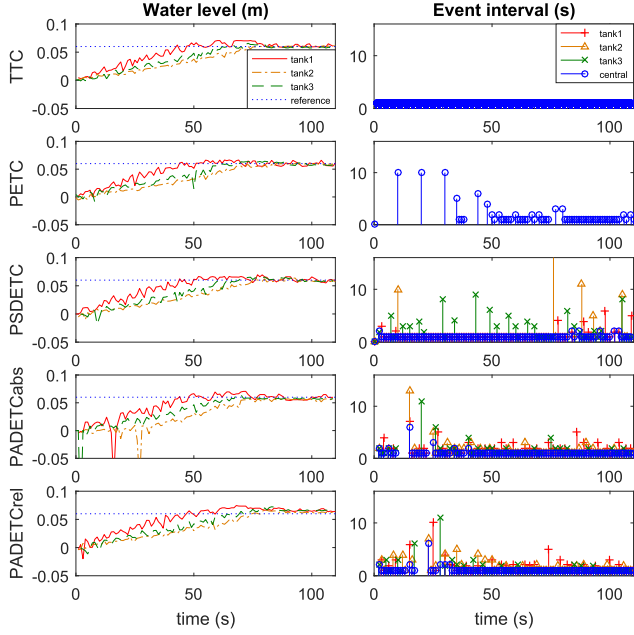


Fig. 9. Experimental results. Note that, for PSDETC, the maximum event interval of small tank 2 is 66 s.

ETC parameters are chosen by finding feasible solutions of the corresponding algorithms (8) and (17), while μ is tuned experimentally. Note that, Assumption 1 holds for all those T listed in Table II.

In the first set of experiments, we examine the impact of the ETC parameters (σ , μ , and ϱ) to the performance of the system. A fixed interval size $T = 1$ was used with all the different combinations of ETC parameter values of Table II. Another set of experiments was conducted to explore the effect of T in the behavior of the system. Keeping $\sigma = 0.2$, $\mu = 0.95$, and $\varrho = 85$ constant, all the experiments were reexecuted with $T = 0.5$ and $T = 2$ [Table II (bold text)]. To ensure the validity of the evaluation results, each experiment was repeated 10 times⁴ for each different combination of ETC parameters and T . Mean values are used to illustrate the evaluation results. The data were captured from the nodes and controller for the period of time between the beginning of each experiment ($t_0 = 0$) until a fixed end time ($t_{\text{end}} = 110$ s), which guarantees the system turns to mode 1 and converges to steady state, denoted T_{exp} .

B. Experimental Results

In this section, we compare TTC, PETC, PSDETC, PADETCabs, and PADETCrel experimental results in terms of

⁴The number of experiment repetitions was selected experimentally by analyzing the variance of the results (i.e., under 2% of mean).

the following.

- 1) *Water Level Overshoot*: the maximum water level during the experiment. This parameter indicates the system's maximum state overshoot which is critical for water network asset safe operation.
- 2) *Switching Time* (t_{sm}): the duration between experiment start time t_0 and the first switch mode time t_{sm} , as described in condition (19). The time to mode switching is employed as an estimate of the system settling time (due to its ease of detection in our setup).
- 3) *Sleep Time*: the total sleep time of all the nodes. This parameter evaluates the use of the bandwidth and CPU in the sensor/actuator node.
- 4) *Discharge (Energy Consumption)*: our custom made sensor module retrieves current measurements c_n in mA at a fixed frequency of 10kHz. The energy consumption of a specific time period Δt in seconds and with average current measurements over this period $\langle C \rangle_{\Delta t}$ can be derived from $E(\Delta t) = \langle C \rangle_{\Delta t} \cdot \Delta t / 3600$. We used a hardware average to ensure the continuity of the current measurements and validated our instrument against a calibrated reference [49]. The energy consumption includes the consumption because of the communication, sensing, actuation, and idle mode. We present two discharge values, i.e., the whole discharge and discharge without sleep time. Based on these parameters, the battery lifetime of different hardware infrastructures can be implied.
- 5) *Actuations*: the number of valves' changes, i.e., $\sum_{t_b \in [t_0, t_{\text{end}}]} \sum_{j=1}^3 \mathcal{Y}_j(t_b)$, where

$$\mathcal{Y}_j(t_b) = \begin{cases} 1, & \text{if } \alpha_j^{\text{in}}(t_b) \neq \alpha_j^{\text{in}}(t_{b-1}) \\ 0, & \text{otherwise} \end{cases}$$

where t_b is defined in (6). The amount of actuations indicates the lifetime of actuators, which is vital for industrial deployments. For example, in water networks, an increase to the valve actuations implies fatigue enhancement of mechanical parts and frequent expensive maintenance.

- 6) *Valve Movement*: the sum of valves' movement in degrees between two consecutive changes, i.e., $\sum_{t_b \in [t_0, t_{\text{end}}]} \sum_{j=1}^3 |\alpha_j^{\text{in}}(t_b) - \alpha_j^{\text{in}}(t_{b-1})|$. Combined with the number of actuations, the valve movement can be used to estimate physical system lifetime.
- 7) *Violations*: the sum of event condition violations. For each violation the local controller transmits a control signal u_j to each node, i.e., three times. Therefore, the total transmitted control signal is equal to three times the violations. This metric indicates the communication requirements of actuators. Violations and actuations are different values, because the local controller can produce the same consecutive control signal.
- 8) *State Transmissions*: the sum of state x_j transmissions to local controller. This metric indicates the communication requirements of sensors. Both violations and state transmissions represents the total communication requirements of the system.

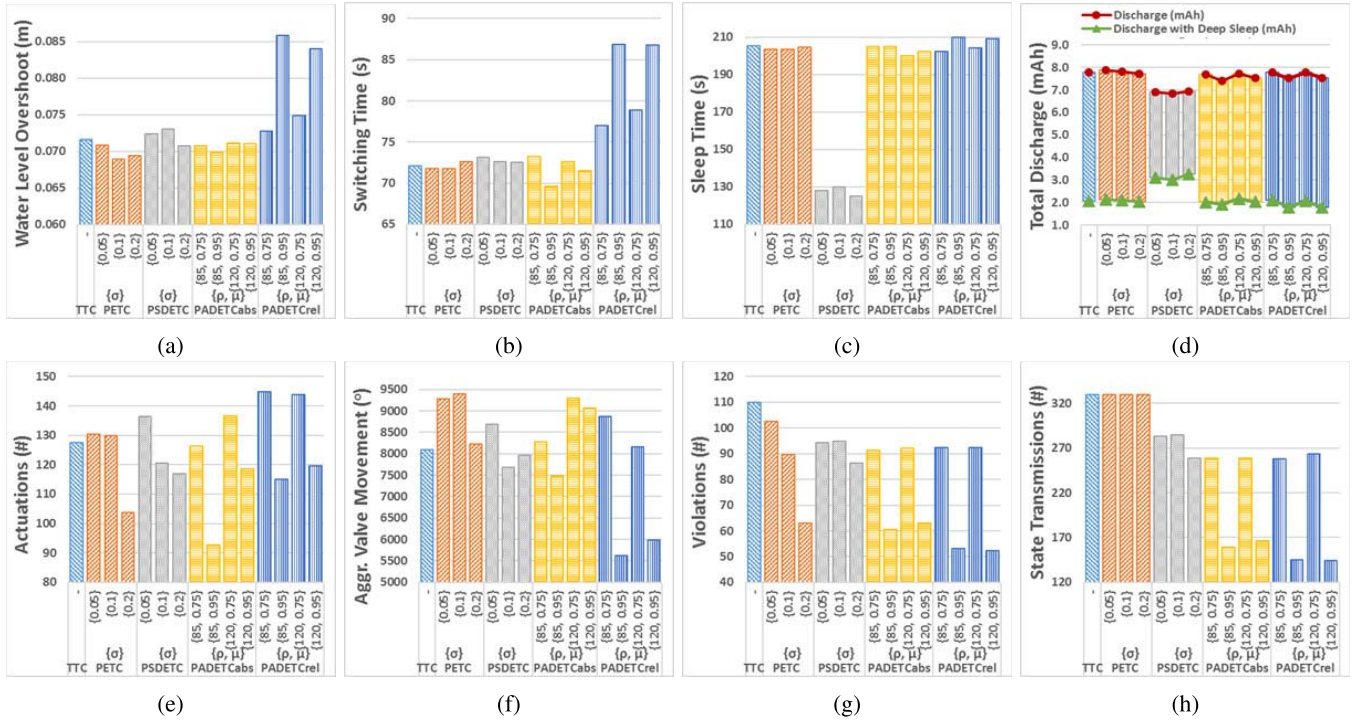


Fig. 10. Impact of ETC parameters (σ , μ , ρ) in: (a) water level overshoot, (b) switching time, (c) sleep time, (d) discharge, (e) actuators, (f) valve movement, (g) violation, and (h) state transmissions.

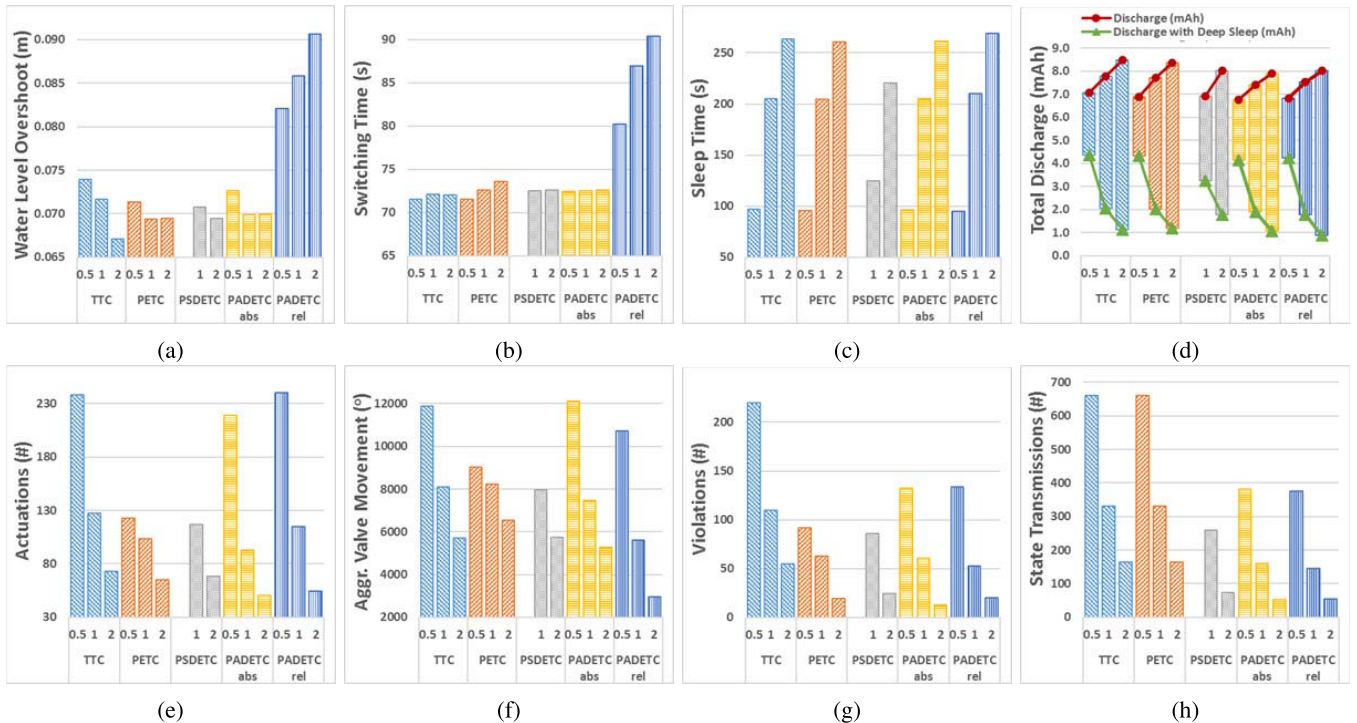


Fig. 11. Impact of interval length T in: (a) water level overshoot, (b) switching time, (c) sleep time, (d) discharge, (e) actuators, (f) valve movement, (g) violation, and (h) state transmissions.

Fig. 9 shows an example of raw data as captured from the nodes and the controller. Next, we analyze the energy consumption trends compared with sleep time for different hardware infrastructures, the effect of ETC parameter setup and interval length T to the performance of the system, and we

aggregate the savings of ETC approaches against the vanilla scenario of TTC.

1) *Energy Consumption and Sleep Time*: The hardware infrastructure of a WaterBox node consumes more energy in sleep mode. During sleep mode, our process yields priority

TABLE III
SAVINGS COMPARED WITH TTC (%) TOTAL EXPERIMENT TIME, I.E., $[0, t_{\text{END}}]$, WITH $\sigma = 0.2$, $\rho = 85$, AND $\mu = 0.95$

Approach	Water Level Overshoot	Switching Time	Discharge	Discharge with Deep Sleep	Actuations	Aggr. Valve Movement	Violations	State Transmissions
PETC	3.18	-0.69	0.97	1.1	18.6	-1.7	42.8	0
PSDETC	1.24	-0.55	11.03	-57.6	8.2	1.5	21.5	21.5
PADETC (abs)	2.44	3.47	4.74	6.9	27.2	7.5	44.8	51.6
PADETC (rel)	-19.72	-20.53	3.28	13.2	9.7	30.5	51.8	56.0

TABLE IV
SAVINGS COMPARED WITH TTC (%) UNTIL MODE SWITCHING TIME, I.E., $[0, t_{\text{SM}}]$, WITH $\sigma = 0.2$, $\rho = 85$, AND $\mu = 0.95$

Approach	Water Level Overshoot	Switching Time	Discharge	Discharge with Deep Sleep	Actuations	Aggr. Valve Movement	Violations	State Transmissions
PETC	3.15	-0.69	1.34	2.4	30.3	17.3	55.2	-0.7
PSDETC	1.67	-0.55	10.97	-66.9	10.4	8.3	14.6	14.6
PADETC (abs)	3.02	3.47	9.22	11.9	34.7	24.1	57.0	63.9
PADETC (rel)	-19.72	-20.53	-15.23	-4.5	14.4	22.6	49.1	54.5

to the background tasks of the operating system which are more energy hungry. In order to generalize the results to different hardware infrastructures which support lower energy consumption during sleep mode (i.e., deep sleep), we provide the upper and a lower bounds of energy consumption. The upper bound presents the real experimental results based on our node while the lower bound represents an estimation of energy consumption of a node which supports deep sleep.⁵ The need of energy consumption range can be clearly seen in Fig. 11(c) and (d). In spite of the sleep time increase in all cases, the upper bound of energy consumption increases proportionally (the opposite holds for the lower bound). Additionally, PSDETC is expected to consume more energy than the others because of the V-slots. However, Fig. 10(c) and (d) illustrate the opposite trend for the upper bound (opposite for lower bound) due to the energy hungry sleep mode. Overall, PADETCabs and PADETCrel consume the least energy compared to the other approaches. In spite of the uses of the same communication scheme, PETC performs slightly better than TTC of actuation reduction (quantitative results will presented later on).

2) *Effect of ETC Parameters:* Fig. 10 presents the effect of different parameters, e.g., σ , ρ , μ with the same interval length $T = 1$. The experiment results follow the trends shown in the theory. In PETC and PSDETC, a smaller σ forces the system to be more conservative and leads to more event condition violations [Fig. 10(g)] and consequently to more actuation [Fig. 10(e)] and energy consumption [Fig. 10(d)]. For the same reason, in the decentralized PSDETC, the state transmission reduces with bigger σ [Fig. 10(h)].

In PADETC, a bigger ρ has similar effect as a smaller σ in PETC. This can be clearly seen in Fig. 10(e) and (g), in which bigger ρ causes more actuations and violations, respectively. A bigger μ can result in more frequent threshold updates, but maintains the threshold less conservative, and thus, the sampling errors can be enlarged. Additionally, Fig. 10(g) shows that μ has greater impact on violations than σ and ρ parameters.

⁵We calculated the lower energy consumption by subtracting the energy consumption during sleep mode from the total

3) *Impact of Interval Length Selection:* Fig. 11 shows the impact of different interval lengths, in which the same predesigned Lyapunov converge rate can be guaranteed, for the same set of rest of the parameters, e.g., σ , ρ , and μ . It can be clearly seen in Fig. 11 that smaller interval T results in better performance but worse energy consumptions. The water level overshoots are almost the same because of the actuator quantization. Larger sampling times always result in longer convergence time and longer sleeping times. Similarly, the upper bound discharge indicates this trend; longer sleep time leads to higher energy consumption due to the energy hungry operating system background tasks. Oppositely, the lower bound of discharge shows that hardware infrastructures with deep sleep consume significant lower energy for larger interval T due to long sleeping time.

4) *Savings Compared with TTC:* Tables III and IV show the total savings of different ETC techniques against TTC for the time period $\text{period}_1 = [0, t_{\text{end}}]$ (total experiment time) and $\text{period}_2 = [0, t_{\text{sm}}]$ (time until switching mode), respectively. We provide this data separately due to the existence of the switched controller and the different behavior of the two modes.

In period_1 PETC performs similar to TTC with the difference of reduced actuations and violations by 18.6% and 42.8%, respectively. In spite the saving, PETC causes more valve movements than TTC. The PSDETC is more conservative than the centralized PETC, with a result, the lower savings in terms of violations. However, PSDETC reduces the valve movements and the state transmissions due to the decentralized architecture. PADETCabs outperforms all the other approaches because of the asynchronous behavior, reducing significantly the violations, state transmissions and actuations by achieving 44.8%, 51.6%, and 27.2% savings, respectively. PADETCrel occurs similar actuation and communication saving with PADETCrel but with the tradeoff of lower performance in terms of water level overshoots and switching time. As has been described in Section II, this happens, because the PADETC with reference value updates introduces an extra error, known as maximum dynamic quantization error. However, this extra error allows this triggering mechanism to

be more robust against noise than any of the other mechanisms with predesigned maximum dynamic error.

In period₂, some ETC approaches deviate compared with the total savings. For example, in PETC approach, period₂ reveals higher actuation savings than period₁. The reason is that in mode 1, the weak pump is unable to supply the tanks with enough water and the system deviates from steady state continuously. Thus, event condition violations are being increased and often large valve movements are required. PSDETC and PADETCabs have a smaller overshoot than the other ETC approaches. Again PADETCabs outperforms the other ETC approaches achieving outstanding violation (57%) and actuation (35%) savings.

VII. CONCLUSION

In this paper, we have proposed duty cycling of the sensing and actuator listening activities and enabled decentralized ETC techniques introducing innovative communication schemes. Specifically, we designed and implemented three new MAC layers, which enable the application of four different PETC and periodic decentralized ETC approaches. By implementing our proposed communication schemes in the WaterBox test bed [35], we provided experimental results from more than 400 experiments.

Based on the experimental results, ETC approaches can introduce considerable benefits into industrial deployments. Due to the outstanding decrease of actuations either in number (up to 35%) or size (i.e., for valve movement up to 24%), the ETC techniques can increase the robustness, resilience, and lifetime of physical plants and actuators significantly. This increase can lead to significant maintenance cost reduction by postponing expensive replacements of plant assets.

WaterBox consists of energy hungry sensor/ actuator nodes to allow computational intensive algorithm deployments. An optimal hardware infrastructure will reduce the energy consumption even more than the evaluation results. Intuitively, the level of energy reduction will be closer to threshold violations (up to 57%) and state transmission (up to 64%) savings which indicate the actuator and sensors communication requirement, respectively.

An additional benefit of applying PETC or periodic decentralized ETC approaches is the reduction of sensing rate. Continuous measurement retrieval from high energy demanding sensors (e.g., the water content sensor [50] which consumes 570 mJ per measurement) may lead to higher energy consumption than the communication process (e.g., low power wide area communication modules in [42] which consumes 1.5 to 42 J per 10 b). Furthermore, based on our experimental results, higher sensing rates do not guarantee higher control performance. As future work, we will examine the aperiodic sensing scheduling over ETC techniques and the impact to the coexisting high sample rate algorithms for anomaly detection and validation. While in this paper, we focus on smart water networks, the proposed framework can be applied to a variety of CPSs, such as smart grids, smart transportation systems, and automated agriculture.

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